

3.CCNP Enterprise作业三——EtherChannel

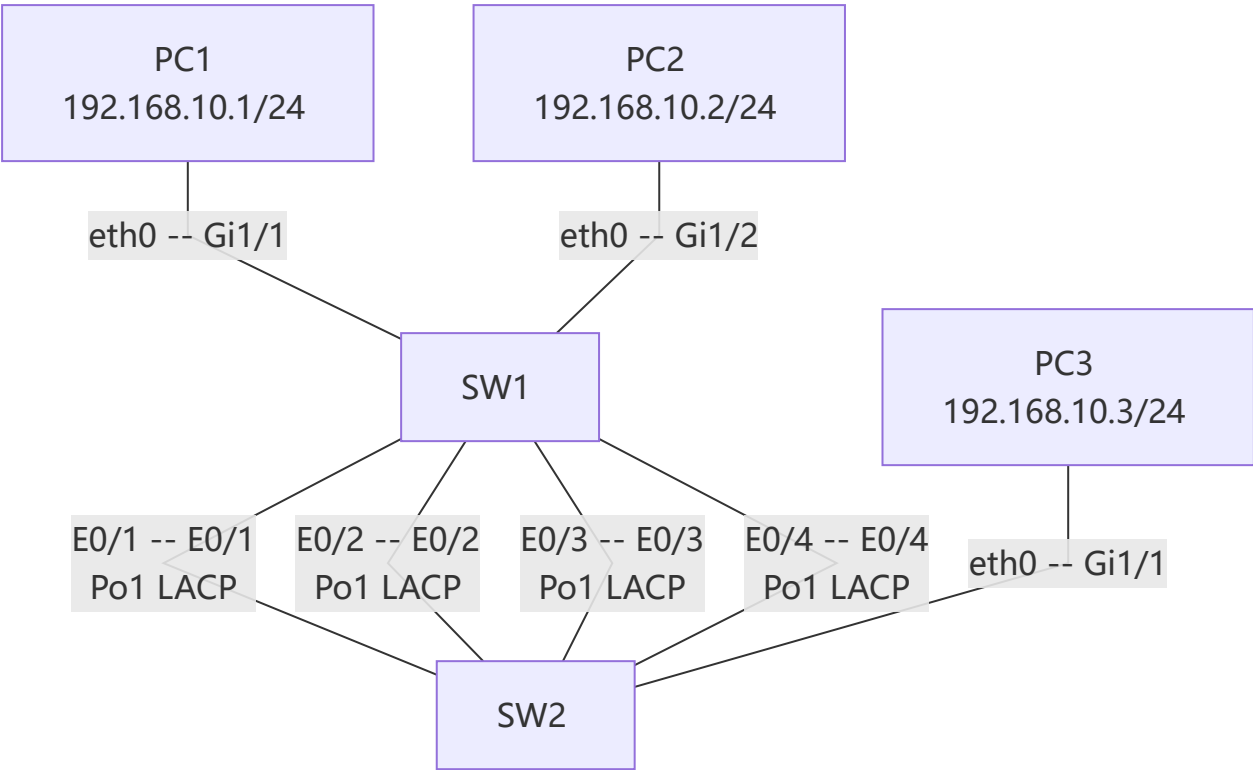
作业主题

EtherChannel

作业目标

- 1. 理解 EtherChannel 的原理与配置

实验拓扑



接口及 IP 地址规划

设备	接口	类型	规划
SW1	E0/1	-	channel-group 1 LACP Active
SW1	E0/2	-	channel-group 1 LACP Active
SW1	E0/3	-	channel-group 1 LACP Active
SW1	E0/4	-	channel-group 1 LACP Active
SW1	Port-Channel 1	Trunk	Allow VLAN 10
SW1	E1/1	Access	VLAN 10

设备	接口	类型	规划
SW1	E1/2	Access	VLAN 10
SW2	E0/1	-	channel-group 1 LACP Passive
SW2	E0/2	-	channel-group 1 LACP Passive
SW2	E0/3	-	channel-group 1 LACP Passive
SW2	E0/4	-	channel-group 1 LACP Passive
SW2	Port-Channel 1	Trunk	Allow VLAN 10
SW2	E1/1	Access	VLAN 10
PC1	eth0	-	192.168.10.1/24
PC2	eth0	-	192.168.10.2/24
PC3	eth0	-	192.168.10.3/24

接口连接规划

设备	接口	连接设备	连接接口
SW1	E0/1	SW2	E0/1
SW1	E0/2	SW2	E0/2
SW1	Gi1/1	PC1	eth0
SW1	Gi1/2	PC2	eth0
SW2	E0/1	SW1	E0/1
SW2	E0/2	SW1	E0/2
SW2	Gi1/1	PC3	eth0

配置需求

1. 请按照所给出的接口连接规划和拓扑搭建实验，请根据接口及 IP 地址规划配置对应的 IP 和接口类型实现 EtherChannel

作业要求

1. 请给出最终配置，并按照设备进行分类
- sw1

```
SW1(config)#int r e0/0-3
SW1(config-if-range)#shutdown
```

```

SW1(config-if-range)#channel-gr
SW1(config-if-range)#channel-group 1 mode ac
SW1(config-if-range)#channel-group 1 mode active ?
SW1(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1

SW1(config-if-range)#no shutd
SW1(config-if-range)#no shutdown
SW1(config-if-range)#exit

SW1(config)#int port-channel 1
SW1(config-if)#sw tr en
SW1(config-if)#sw tr encapsulation d
SW1(config-if)#sw tr encapsulation dot1q
SW1(config-if)#sw mo tr

SW1(config-if)#sw tr allowed vlan all
SW1(config-if)#end
SW1#
*Sep 11 10:11:45.505: %SYS-5-CONFIG_I: Configured from console by console
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int r e1/0-1
SW1(config-if-range)#sw mo ac
SW1(config-if-range)#sw ac vlan 10
% Access VLAN does not exist. Creating vlan 10

SW1#show etherchannel summary
Flags:  D - down          P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3        S - Layer2
        U - in use        N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)          LACP        Et0/0(P)   Et0/1(P)   Et0/2(P)
                                   Et0/3(P)

```

```
SW1#show in po
% Ambiguous command: "show in po"
SW1#show in
SW1#show int
SW1#show interfaces por
SW1#show interfaces port-c
SW1#show interfaces port-channel 1
Port-channel1 is up, line protocol is up (connected)
  Hardware is EtherChannel, address is aabb.cc00.0100 (bia aabb.cc00.0100)
  MTU 1500 bytes, BW 40000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, Auto-speed, media type is RJ45
  input flow-control is off, output flow-control is unsupported
  Members in this channel: Et0/0 Et0/1 Et0/2 Et0/3
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:42, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/2000/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 2000 bits/sec, 1 packets/sec
    6 packets input, 297 bytes, 0 no buffer
    Received 6 broadcasts (0 multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 input packets with dribble condition detected
    168 packets output, 28592 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
SW1#
```

sw2

```
W2(config)#int e1/0
SW2(config-if)#sw mo ac
SW2(config-if)#sw ac vl 10
% Access VLAN does not exist. Creating vlan 10
SW2(config-if)#int r e0/0-3
SW2(config-if-range)#shutdown

SW2(config-if-range)#cha
SW2(config-if-range)#channel-gr
SW2(config-if-range)#channel-group 1 mode ac
SW2(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1

SW2(config-if-range)#no shut
```

```

SW2(config-if-range)#no shutdown
SW2(config-if-range)#exit
ype.
SW2(config)#int port-c
SW2(config)#int port-channel
*Sep 11 10:13:22.252: %LINK-3-UPDOWN: Interface Port-channel1, changed state to up
*Sep 11 10:13:23.256: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed
state to up
SW2(config)#int port-channel 1
SW2(config-if)#sw tr en do
SW2(config-if)#sw mo
*Sep 11 10:13:29.273: %LINK-3-UPDOWN: Interface Port-channel1, changed state to down
*Sep 11 10:13:30.274: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed
state to down
SW2(config-if)#sw mo tr
SW2(config-if)#sw tr al vl all
SW2(config-if)#
*Sep 11 10:13:34.796: %LINK-3-UPDOWN: Interface Port-channel1, changed state to up
*Sep 11 10:13:35.796: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed
state to up
SW2(config-if)#end
SW2#show
*Sep 11 10:13:41.378: %SYS-5-CONFIG_I: Configured from console by console
SW2#show eth
SW2#show etherc
SW2#show etherchannel sum
SW2#show etherchannel summary
Flags: D - down          P - bundled in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        N - not in use, no aggregation
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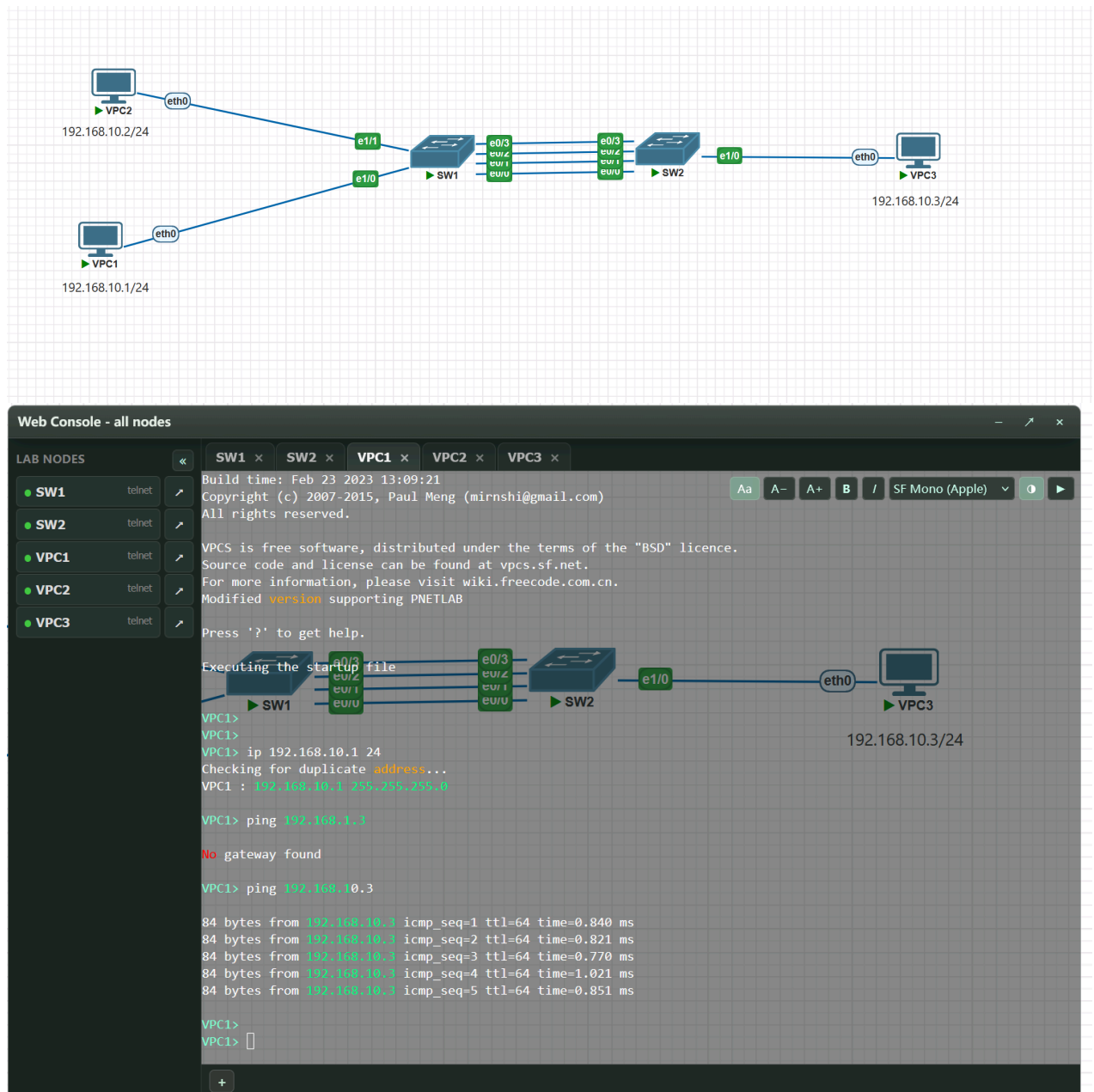
       A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Et0/0(P)   Et0/1(P)   Et0/2(P)
                                   Et0/3(P)
SW2#

```

2. 请给出连通性验证截图



3. 请比较 src-dst-ip 与 src-dst-mac 的负载均衡方式，请详细论述

src-dst-ip:源 MAC + 目的 MAC进行hash，防止单一字段，流量全从一条链路上走

src-dst-mac:源 IP + 目的 IP进行hash，成员链路更加均衡